BEC Hamilton County Conversion Project – ERCOT Independent Review Study Status Update

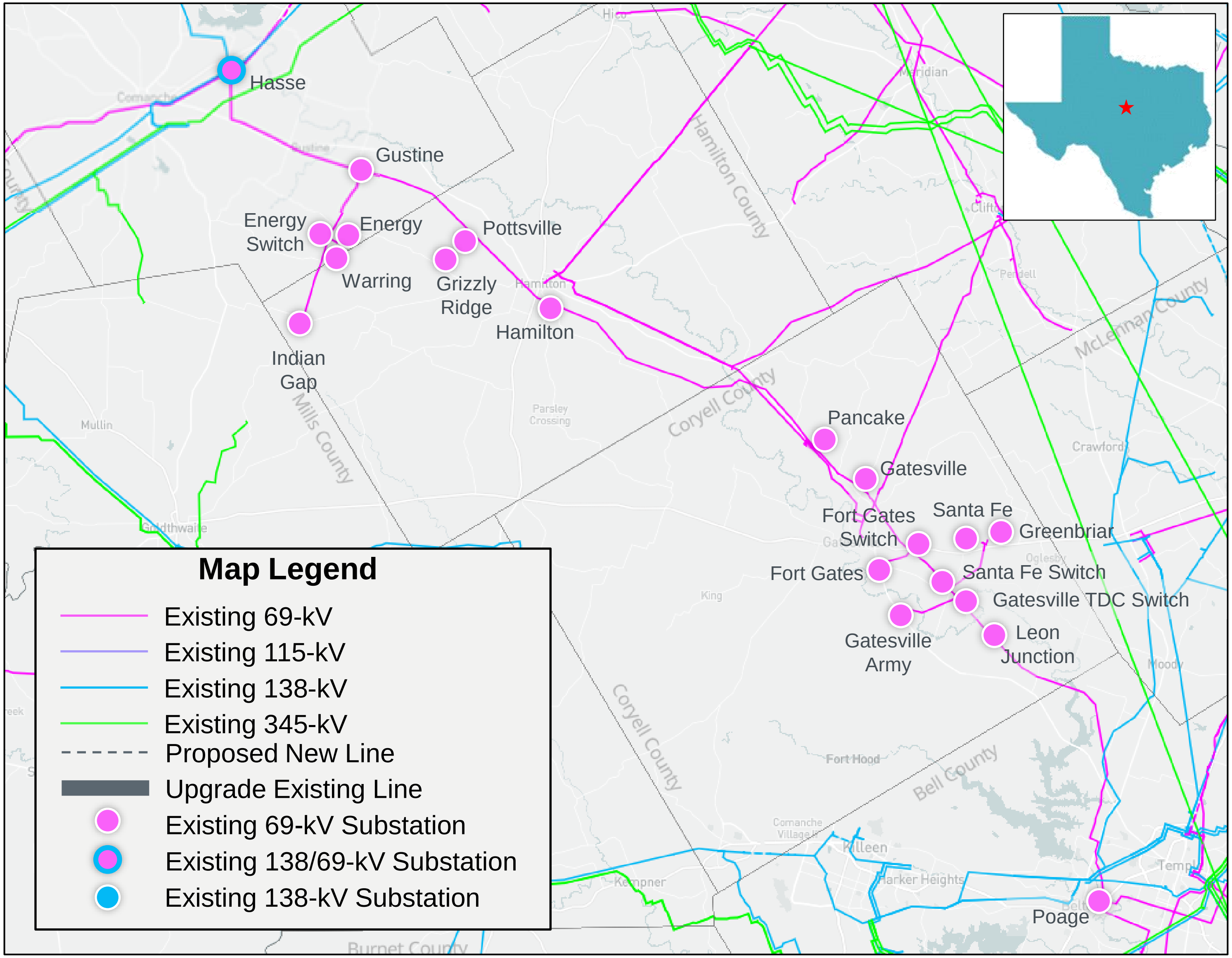
Travis Head

RPG Meeting
June 17, 2025

Introduction

- BEC submitted the Hamilton County Conversion Project for Regional Planning Group (RPG) review in February 2025
  - This Tier 2 project is estimated at $90.0 million and will require a Convenience and Necessity (CCN)
  - Estimated in-service date (ISD) is Fall 2030
  - Is a GTC Exit Strategy for the Hamilton GTC
- This project is currently under ERCOT Independent Review (EIR)

Study Area Map



Study Assumptions – Base Case

- Study Area
  - North Central Weather Zone, focusing on transmission in Hamilton, Comanche, Coryell, and Bell Counties
  - Monitor surrounding counties that are electrically close to the area
- Steady-State Base Case
  - Final 2024 Regional Transmission Planning (RTP) 2029 summer peak case will be updated to construct the study base case posted in Market Information System (MIS)
    - Case: 2024RTP_2030_SUM_12202024
    - Link: https://mis.ercot.com/secure/data-products/grid/regional-planning

Study Assumptions – Load, Reserve, Transmission, & Generation

- Load in study area
  - ~43 MW of Officer Letter load was added to the study area
- Reserve
  - Reserve levels are consistent with the 2024 RTP
- Transmission
  - See Appendix A for a list of transmission projects added
  - See Appendix B for a list of RTP placeholder projects that were removed
- Generation
  - See Appendix C for a list of generation projects added

Study Assumptions – Economic Study

- 2024 RTP economic study base case for 2029 will be used for this study
- 2024 RTP economic study assumptions will be maintained in this study
  - Financial Assumptions in February 2024 RTP meeting
    - https://www.ercot.com/calendar/02122024-RPG-Meeting
  - Economic Assumptions in March 2024 RTP meeting
    - https://www.ercot.com/calendar/03182024-RPG-Meeting-_-Webex
  - Stability Interface Limits in July 2024 RTP meeting
    - https://www.ercot.com/calendar/07162024-RPG-Meeting

Preliminary Results of Reliability Assessment – Base Case

Contingency Category*	Unsolved Power Flow	Voltage Violations	Thermal Overloads
P1	None	None	None
P3 (G-1+N-1)*	None	None	None
P6.2 (X-1+N-1)**	None	None	None
P7	None	None	None

*G-1 Generators tested: Grizzly Ridge Solar S1, Panda Temple C1, Logans Gap Wind W1
**X-1 Transformers tested: Hasse X1, Comanche Switch X1, Temple Switch X1
***Violations seen in the basecase under P1 events were also seen under G-1 and X-1 events

Option 1 – BEC Proposed Project

- Convert existing Hasse to Gustine 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 418 MVA or greater, approximately 14.0-mile;
- Convert the existing Gustine, Energy, Energy Switch, Waring, Indian Gap, Pottsville, Hamilton, Gatesville, Fort Gates Switch, Fort Gates, Gatesville TDC Switch, Leon Junction, Poage 69-kV substations to 138-kV operation;
- Rebuild existing Gustine to Energy Switch 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 5.1-mile;
- Rebuild existing Energy Switch to Energy 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 1.0-mile;
- Rebuild existing Energy Switch to Indian Gap 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 6.6-mile;

Option 1 – BEC Proposed Project cont'd

- Convert existing Gustine to Pottsville 69-kV transmission line to operate a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 11.0-mile;
- Convert existing Pottsville to Hamilton 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 7.6-mile;
- Convert existing Pottsville to Grizzly Ridge 69-kV transmission line to operate as a 138-kV transmission line with normal rating of 220 MVA and emergency rating of 242 MVA or greater, approximately 0.1-mile;
- Install two 138/69-kV autotransformers at the existing Pancake 69-kV substation with normal and emergency ratings of 60 MVA;
- Convert existing Hamilton to Pancake 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 21.9-mile;
- Convert existing Pancake to Gatesville 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 4.0-mile;

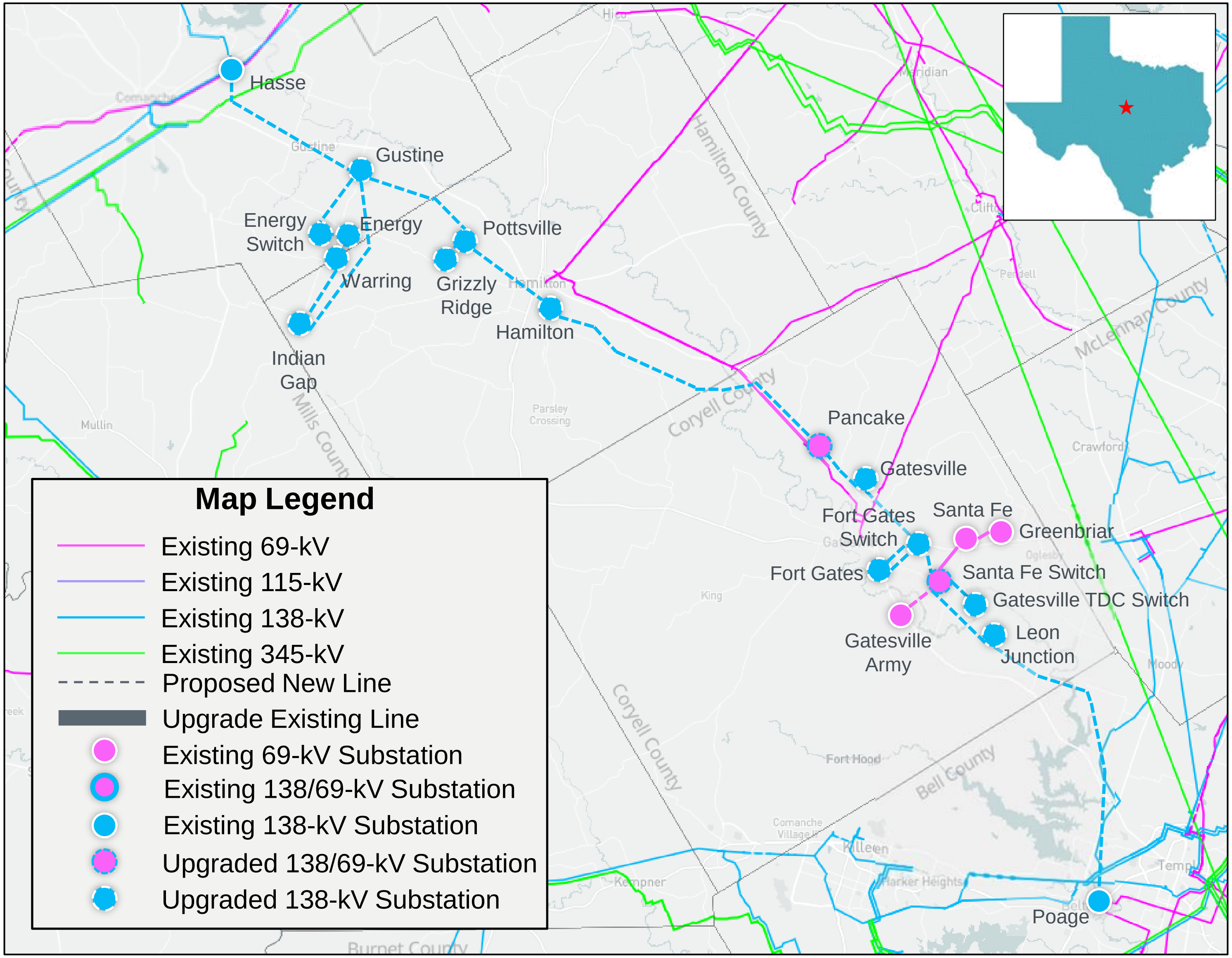
Option 1 – BEC Proposed Project cont'd

- Convert existing Gatesville to Fort Gates 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 7.5-mile;
- Rebuild existing Fort Gates Switch to Fort Gates 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 2.0-mile;
- Convert existing Fort Gates to Santa Fe Switch 69-kV transmission line as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 3.9-mile;
- Install two 138/69-kV autotransformers at the existing Santa Fe Switch 69-kV with normal and emergency ratings of 60 MVA;
- Rebuild existing Santa Fe Switch to Gatesville TDC Switch 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 0.8-mile;
- Move existing Gatesville TDC Switch to Gatesville Army 69-kV tie line to connect to Santa Fe Switch creating the Santa Fe Switch to Gatesville Army 69-kV tie line;

Option 1 – BEC Proposed Project cont'd

- Convert existing Santa Fe Switch to Leon Junction 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 4.0-mile;
- Convert existing Leon Junction to Poage 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 24.4-mile; and
- Retire Poage autotransformer (Installed from Temple Area Improvements RPG).

Option 1 – BEC Proposed Project



Option 2 – ERCOT GTC Exit Plan

- Convert the existing Gustine, Energy, Energy Switch, Waring, Indian Gap, Pottsville, and Hamilton 69-kV BEC owned substations to 138-kV operation;
- Remove existing 138/69-kV transformers at Hasse;
- Convert existing Hasse to Gustine 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 418 MVA or greater, approximately 14.0-mile;
- Convert existing Gustine to Energy Switch 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 524 MVA or greater, approximately 5.1-mile;
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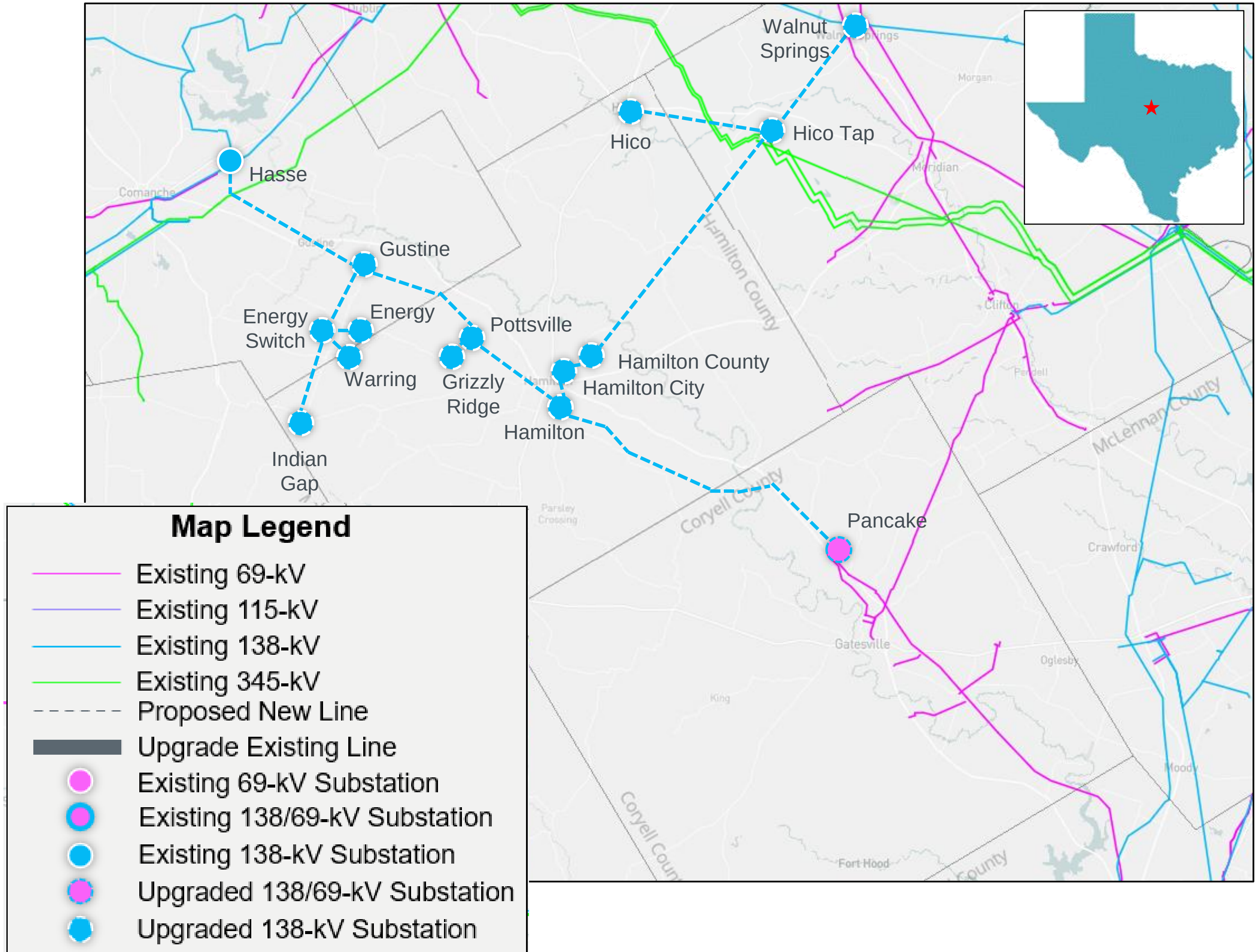
- Convert existing Hamilton to Pancake 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 21.9-mile;
- Convert the existing Hamilton City, Hamilton County, Hico, Hico Tap, and Walnut Springs 69-kV TNMP owned substations to 138-kV operation;

Option 2 – ERCOT GTC Exit Plan cont’d

- Construct a new Hamilton to Hamilton County 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 2.0-mile;
- Convert existing Hamilton City to Hamilton County 69-kV transmission line to operate a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 0.9-mile;
- Convert existing Hamilton County to Hico Tap 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 22.3-mile;
- Convert existing Hico Tap to Walnut Springs 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 10.5-mile;
- Convert existing Hico Tap to Hico 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 9.9-mile; and
- De-energize the existing Hamilton County to Jonesboro 69-kV transmission line.

Option 2 – ERCOT GTC Exit Plan



Option 3 – Combined Option

- Convert the existing Gustine, Energy, Energy Switch, Waring, Indian Gap, Pottsville, and Hamilton 69-kV BEC owned substations to 138-kV operation;
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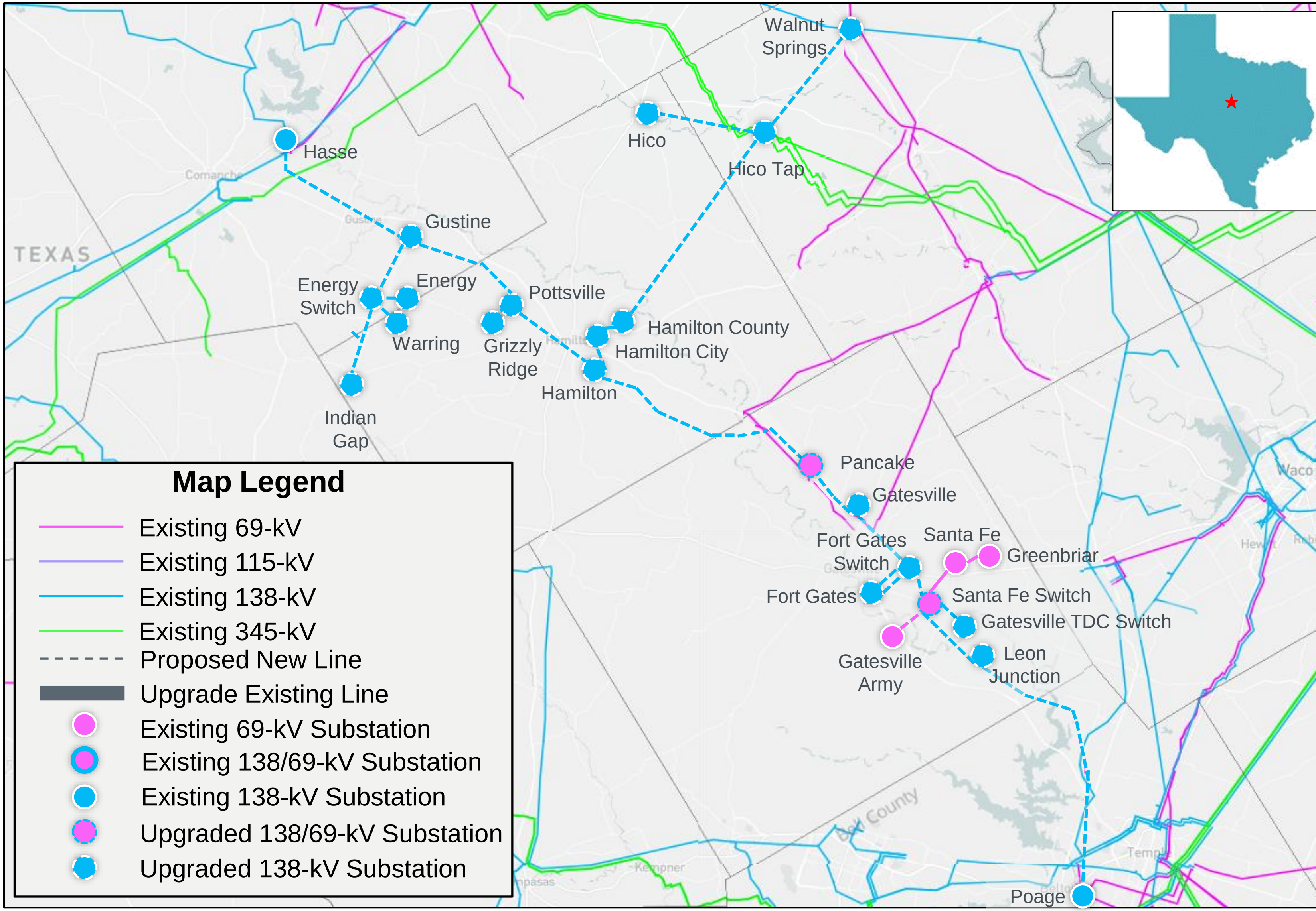

Economic Evaluation of Options

Option	Total Cost	Annual Cost Passing Requirement %	Consumer Revenue Passing Requirement %	Benefit to Cost Ratio (Annual Production Cost) %	Benefit to Cost Ratio (Consumer Revenue) %
O1 – BEC	90 $M	13.0 %	12.7 %	0.3 %	16.9 %
O2 – ERCOT	110 $M	13.0 %	12.7 %	0.9 %	15.0 %
O3 – Combined	~ 150 $M	13.0 %	12.7 %	2.1 %	9.8 %

Study Procedure

- Project Evaluation
  - Additional project alternatives may be tested to satisfy the NERC and ERCOT reliability requirements
  - ERCOT may also perform the following studies
    - Planned maintenance outage
    - Long-Term Load-Serving Capability Assessment
  - ERCOT protocols defined in Protocol 3.11.2(5) will be followed to perform the production cost savings test and consumer revenue reduction test.
- Congestion Analysis
  - Congestion analysis may be performed based on the recommended transmission upgrades to ensure that the identified transmission upgrades do not result in new congestion within the study area

Deliverables

- Tentative Timelines
  - Status updates at the future RPG meetings
  - Final Recommendation – Q3 2025

Thank you!

Stakeholder comments also welcomed through:

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Appendix A – Transmission Projects Added

TPIT/RPG No	Project Name	Tier	Project ISD	County(s)
78173	BEPC_27TPIT78173_Trimmier_Capacitor	Tier 4	3/15/2027	Bell
78179	BEPC_27TPIT78179_DingDong_Capacitor	Tier 4	3/15/2027	Bell

Appendix B – Transmission Backed Out

RTP Project ID	Project Name	County(s)
2024-NC12	Knob Creek Switch (3413) to Salado Switch (3699) 345-kV Line Upgrade and Substation Rebuilds	Bell
2024-NC37	Bell East (3687) to Salado Switch (3699) 345-kV Line Addition and Substation Rebuilds	Bell, Williamson
2024-NC43	Temple Switch (3415) to Belton (3610) 138-kV Line Upgrades	Bell
2024-NC60	Bell County East Switch (3687) to Littlepond (3377) and Bell County East Switch (3687) to Brangus Switch (3705) 345-kV Line Upgrades and Substation Rebuilds	Milam, Bell
2024-NC92	Killeen (3423) 138-kV Cap Bank Addition	Bell

Appendix C – Generation Added

GINR	Project Name	Fuel	Project COD	Capacity (~MW)	County
23INR0079	Chillingham Storage	OTH	5/1/2025	153.9	Bell
23INR0249	Limewood Solar	SOL	12/31/2025	204.6	Bell
23INR0344	Hermes Solar	SOL	9/30/2025	100.4	Bell
23INR0469	Big Elm Storage	OTH	8/15/2026	100.8	Bell
24INR0166	Stillhouse Solar	SOL	9/1/2025	210.8	Bell
24INR0365	Hermes Storage	OTH	9/30/2025	100.4	Bell
22INR0220	Lamkin Solar	SOL	8/8/2027	100.5	Hamilton
24INR0181	Bynum Solar	SOL	10/1/2025	56	Hamilton
25INR0100	Goldeneye BESS	OTH	4/1/2027	201.6	Bell
22INR0596	Grizzly Ridge BESS	OTH	2/10/2023	9.95	Hamilton